



CHRIST
(DEEMED TO BE UNIVERSITY)
BANGALORE · INDIA

BUNDLEWISE: A RETAIL ANALYTICS TOOL FOR SMART BUNDLING AND PROFIT-DRIVEN PRICING

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Presented To

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Guide

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Mission

Christ University is a nurturing ground for an individual's holistic development to make effective contribution to the society in a dynamic environment

Vision

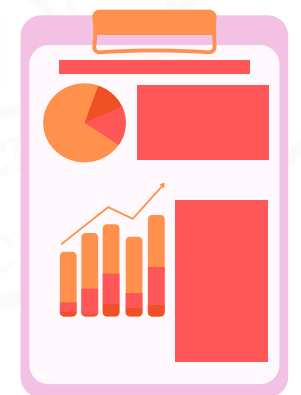
Excellence and Service

Core Values

Faith in God | Moral Uprightness
Love of Fellow Beings | Social Responsibility
Pursuit of Excellence

The Retailer's Challenge

- **Problem:** Small to mid-size retailers often struggle to:
 - Understand complex raw sales data.
 - Identify hidden purchasing patterns.
 - Predict future trends or optimize pricing.
 - Translate data into actionable business decisions.
- **Goal:** Provide an accessible, powerful tool to transform raw data into clear, actionable insights.

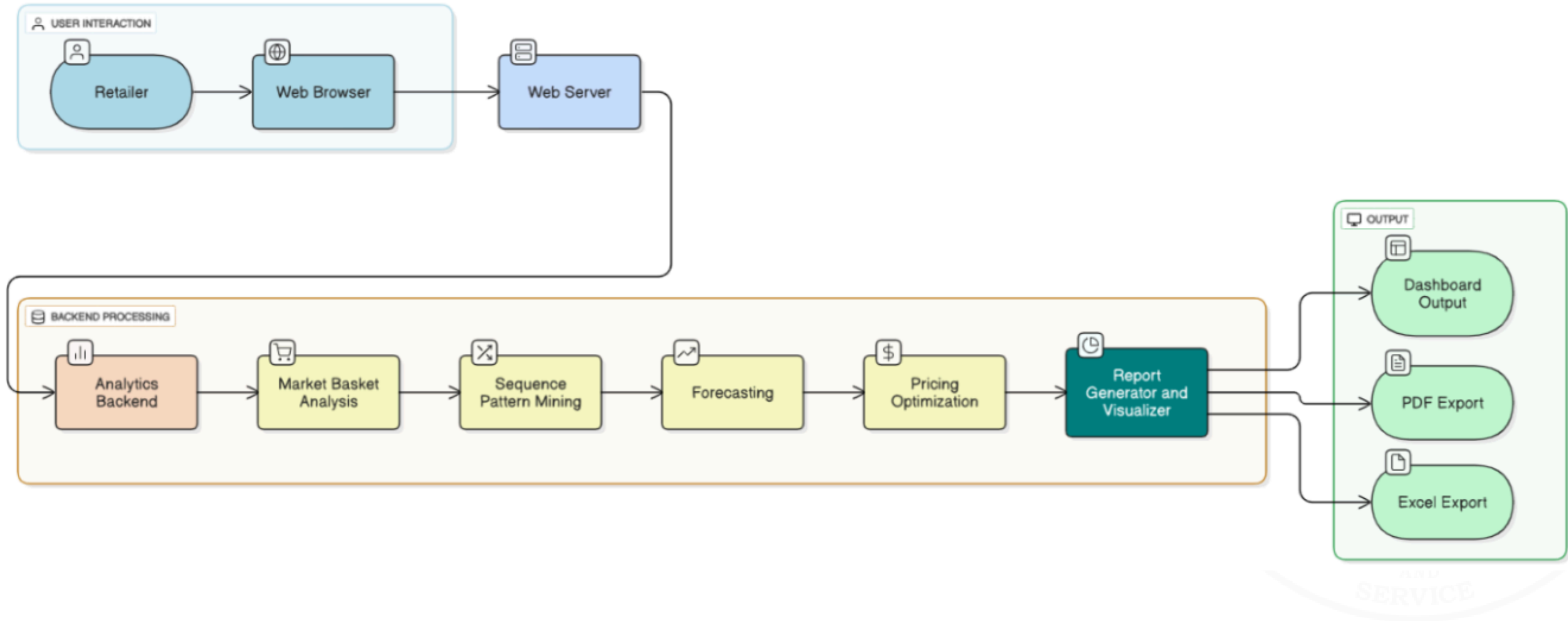


Our Solution

- **Vision:** A user-friendly web application designed to empower retailers by:
 - Simplifying data ingestion and cleaning.
 - Automating advanced sales analysis (what's bought together, what's bought next).
 - Generating clear, actionable recommendations.
- **Key Pillars:** Data Quality, Pattern Discovery, Strategic Recommendations.



End-to-End Project Flow



Progress So Far: Laying the Foundation

- **Data Ingestion & Preprocessing Module**

- **Robust File Upload:** Supports CSV, XLSX, XLS formats.
- **Automated Data Cleaning:** Handles missing critical data, negative quantities/prices, and duplicate entries.
- **Smart Validation:** Validates schema and data types, standardizes column names (e.g., 'invoiceno' to 'Transaction ID').
- **Data Persistence:** Processed data is stored in memory, eliminating need for re-upload on navigation.





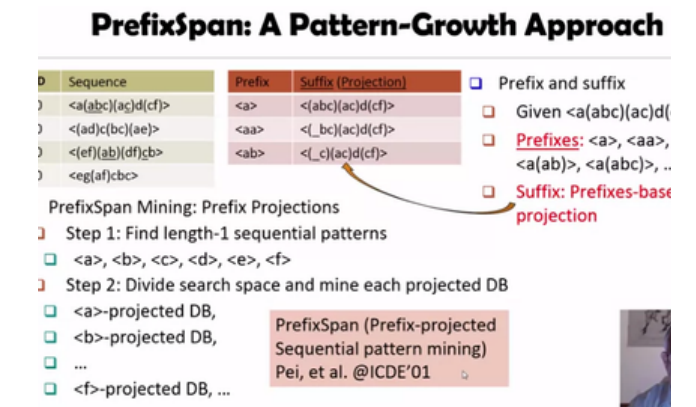
Progress So Far: Core Analytics - Market Basket Analysis (MBA)



- **Purpose:** Discovering frequently bought product bundles (e.g., "customers who buy A often buy B").
- **Algorithm:** Implemented the Eclat algorithm for efficient frequent itemset mining.
- **Key Metrics:** Calculates and displays Support, Confidence, and Lift for each association rule.
- **Enhanced Insights:** Curates and highlights "Top Actionable Insights" by prioritizing unique, high-Lift/Confidence rules (avoiding redundant permutations).



Progress So Far: Core Analytics - Sequential Pattern Mining (SPM)



- **Purpose:** Uncovering common sequences of purchases over time (e.g., "customer buys A, then B, then C").
- **Algorithm:** Implemented a custom PrefixSpan algorithm for mining ordered patterns from customer journeys.
- **Key Metric:** Calculates and displays Support for sequential patterns.
- **Actionable Phrases:** Presents patterns as human-readable insights (e.g., "After buying X, customers often buy Y next.").



User-Centric Features & Improved UX



- **Intuitive Parameter Suggestions:** "Suggest Support" button provides data-driven, rounded starting points for min_support in both MBA and SPM, simplifying complex thresholds for users.
- **Unified Dashboard:** A new dedicated "Dashboard" page provides an immediate overview, automatically running analyses with suggested parameters and displaying key visualizations side-by-side.
- **Consistent & Aligned UI:** Streamlined design, tabbed navigation, and clear, standardized result displays across all modules (Data Ingestion, MBA, SPM, Dashboard).
- **Seamless Navigation:** Easy "Back to Data Ingestion" links, with processed data remembered.

Demo: Data Upload Page

BundleWise - Data Ingestion

127.0.0.1:5000/data/

Retail Analytics Platform

Transform Your Sales Data Into Insights

Upload your retail dataset to get comprehensive analytics including market basket analysis, sales forecasting, and pricing optimization recommendations.

Upload Sales Dataset

Upload your CSV or Excel file containing transaction data to begin analysis.

Drag & drop your file
or click to browse files

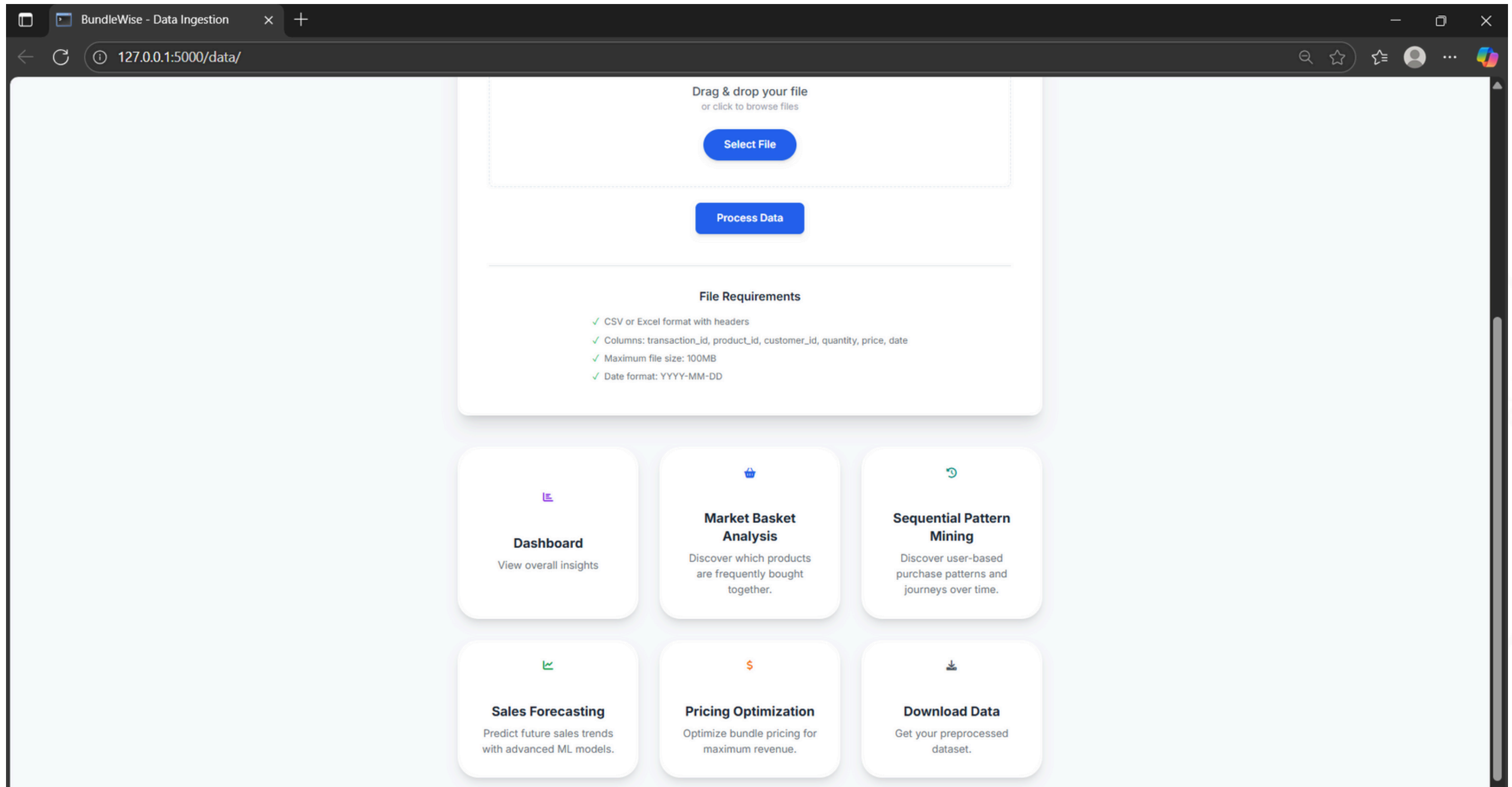
Select File

Process Data

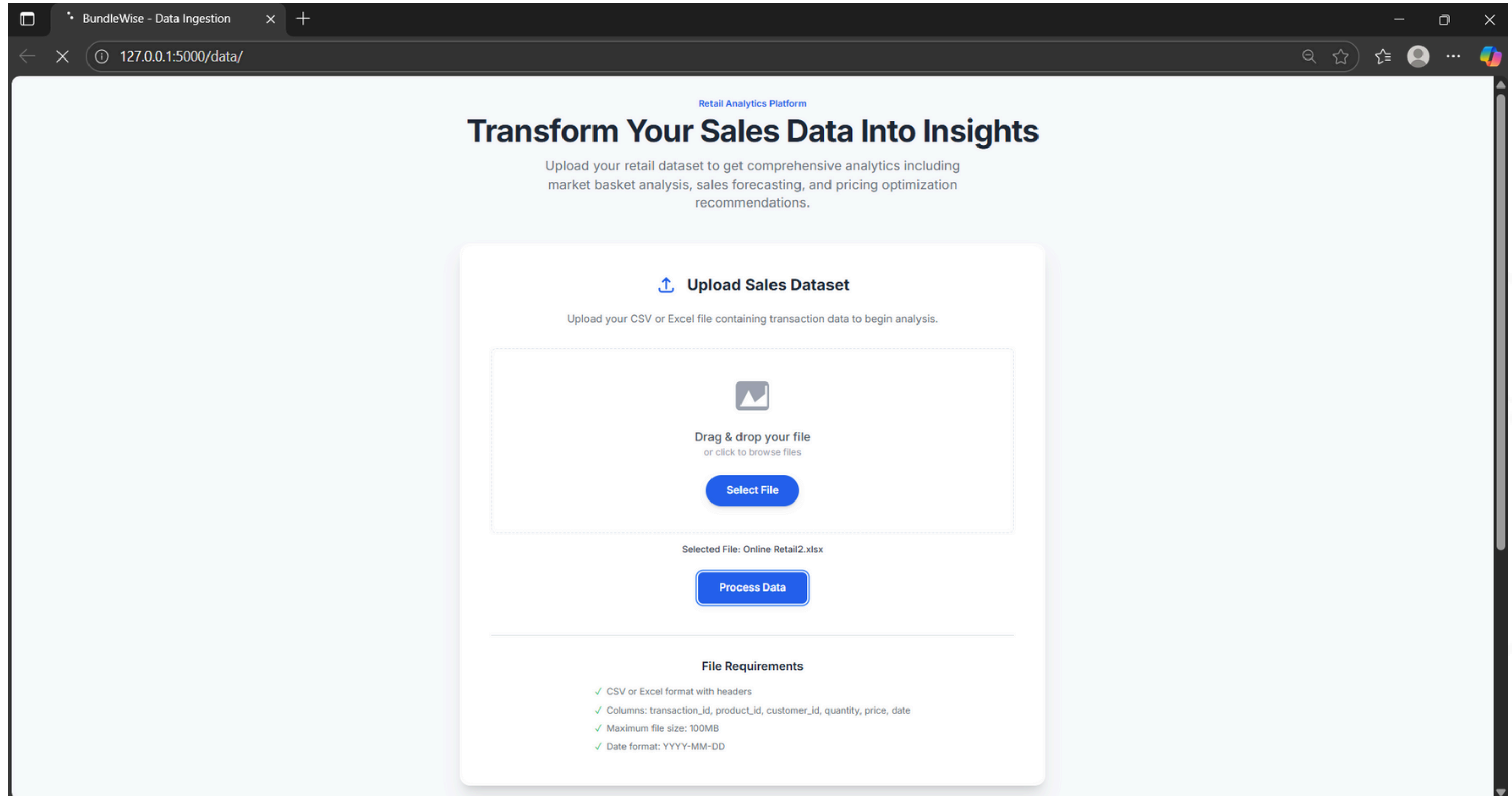
File Requirements

- ✓ CSV or Excel format with headers
- ✓ Columns: transaction_id, product_id, customer_id, quantity, price, date
- ✓ Maximum file size: 100MB
- ✓ Date format: YYYY-MM-DD

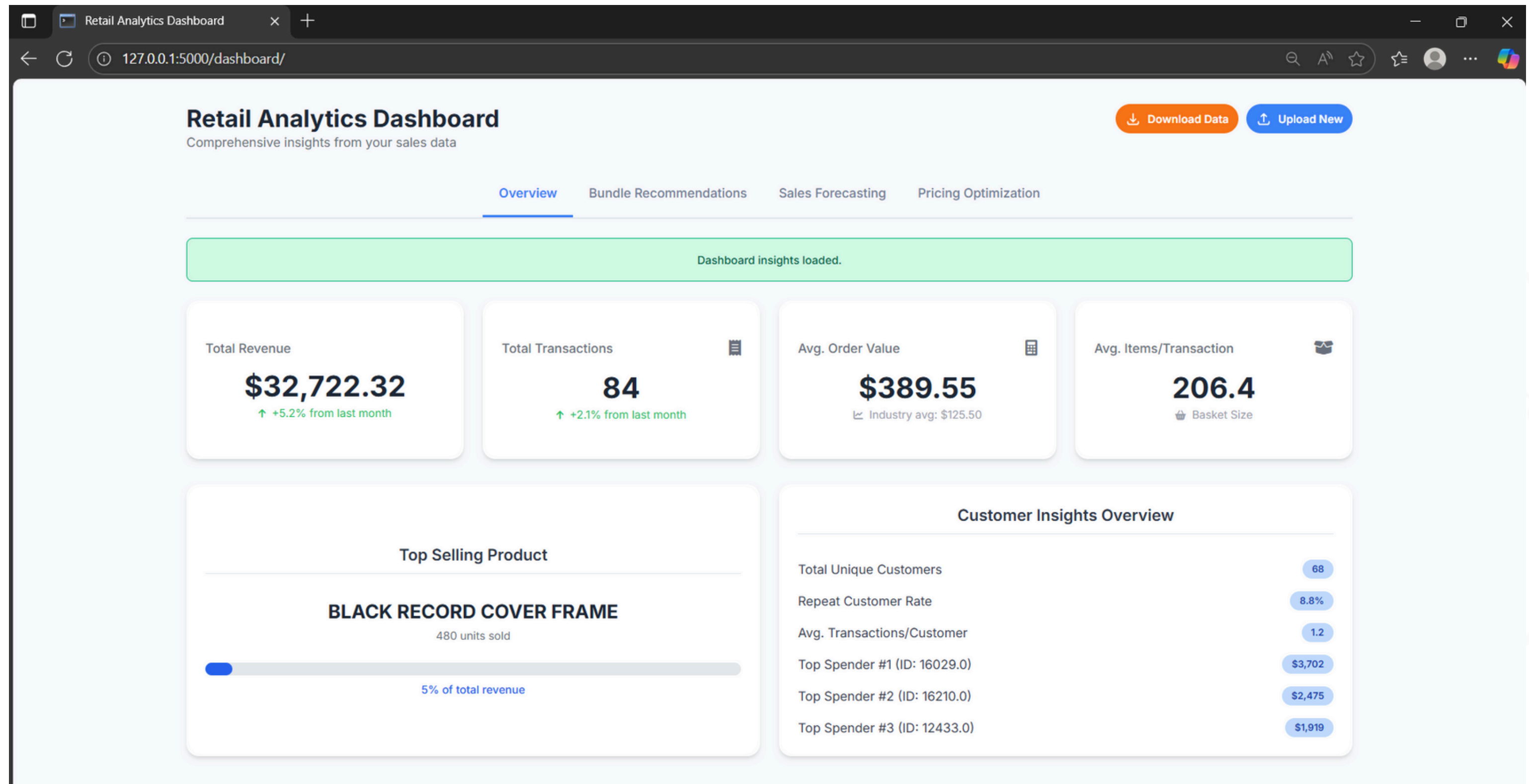
Demo: Data Upload Page



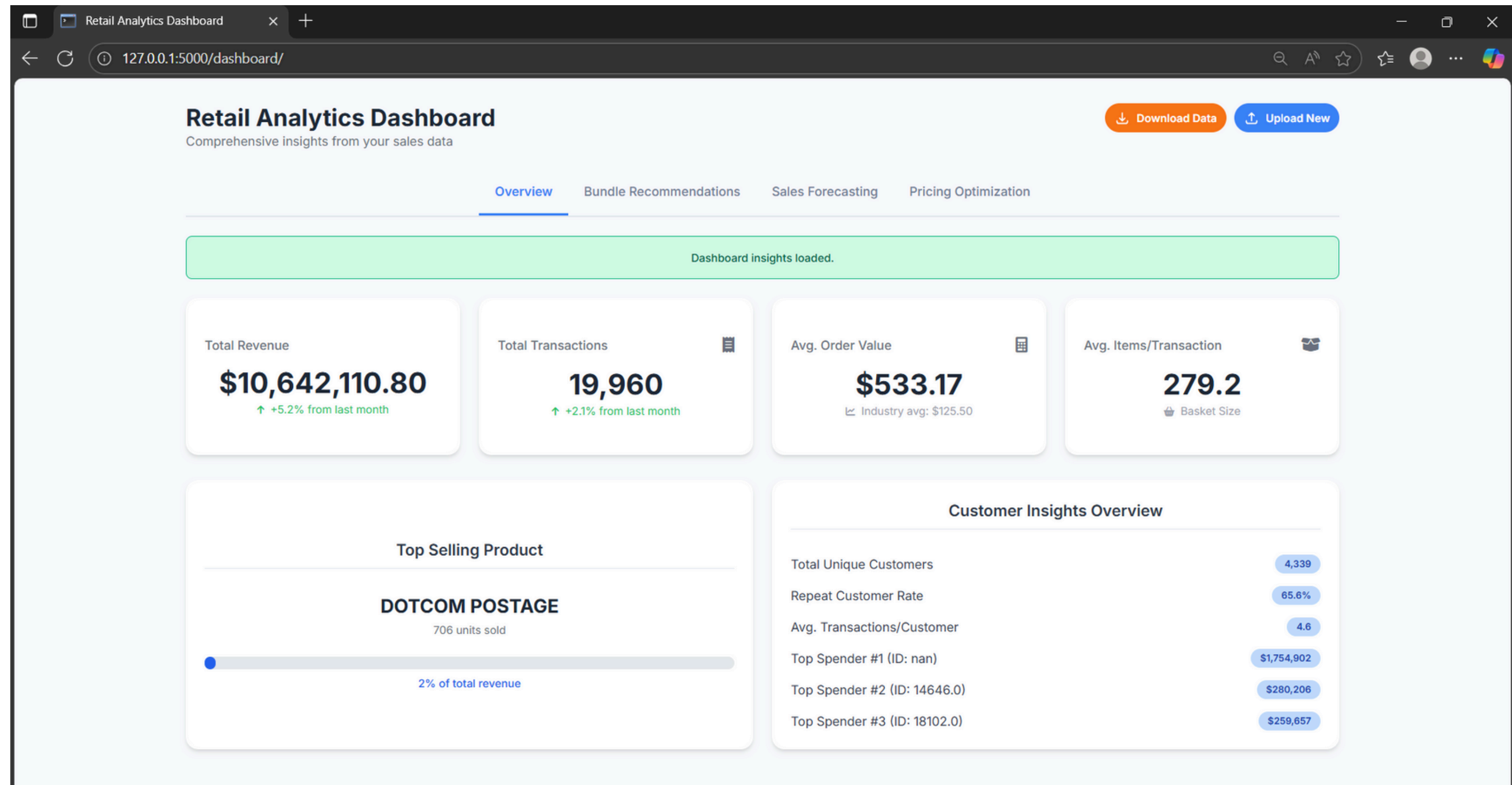
Demo: Data Uploaded and Processed



Demo: Overview of Data in Dashboard



Demo: Overview of Data in Dashboard



Demo: Bundle Recommendations Tab (Displays Bundles)

Retail Analytics Dashboard

127.0.0.1:5000/dashboard/

Retail Analytics Dashboard

Comprehensive insights from your sales data

Download Data

Upload New

Overview

Bundle Recommendations

Sales Forecasting

Pricing Optimization

This tab displays recommended bundles based on Market Basket Analysis and Sequential Pattern Mining.

Actionable Bundle Recommendations

"Buy 1 Get 1 Free" Offers (from MBA)

Buy: VINTAGE BILLBOARD LOVE/HATE MUG

Get Free: WOODEN PICTURE FRAME WHITE FINISH

Customers buying VINTAGE BILLBOARD LOVE/HATE MUG are highly likely to also buy WOODEN PICTURE FRAME WHITE FINISH (Conf: 100.0%, Lift: 21.0x).

Buy: LOVE BUILDING BLOCK WORD

Get Free: HOME BUILDING BLOCK WORD

Customers buying LOVE BUILDING BLOCK WORD are highly likely to also buy HOME BUILDING BLOCK WORD (Conf: 100.0%, Lift: 21.0x).

Buy: WHITE METAL LANTERN

Get Free: GLASS STAR FROSTED T-LIGHT HOLDER

Customers buying WHITE METAL LANTERN are highly likely to also buy GLASS STAR FROSTED T-LIGHT HOLDER (Conf: 100.0%, Lift: 16.8x).

Buy: CREAM CUPID HEARTS COAT HANGER

Get Free: GLASS STAR FROSTED T-LIGHT HOLDER

Customers buying CREAM CUPID HEARTS COAT HANGER are highly likely to also buy GLASS STAR FROSTED T-LIGHT HOLDER (Conf: 100.0%, Lift: 16.8x).

Buy: WHITE METAL LANTERN

Get Free: CREAM CUPID HEARTS COAT HANGER

Customers buying WHITE METAL LANTERN are highly likely to also buy CREAM CUPID HEARTS COAT HANGER (Conf: 100.0%, Lift: 16.8x).

"Buy X, Get Y% Off Y" Offers (from SPM)

Buy: JAM MAKING SET PRINTED

Next, Buy: HAND WARMER RED POLKA DOT

Popular sequential purchase recommendation based on customer patterns.

Buy: HAND WARMER SCOTTY DOG DESIGN

Next, Buy: PAPER CHAIN KIT 50'S CHRISTMAS

Popular sequential purchase recommendation based on customer patterns.

Buy: PAPER CHAIN KIT VINTAGE CHRISTMAS

Next, Buy: HAND WARMER OWL DESIGN

Popular sequential purchase recommendation based on customer patterns.

Buy: ASSORTED COLOUR BIRD ORNAMENT

Next, Buy: HAND WARMER BIRD DESIGN

Popular sequential purchase recommendation based on customer patterns.

Buy: PACK OF 60 PINK PAISLEY CAKE CASES

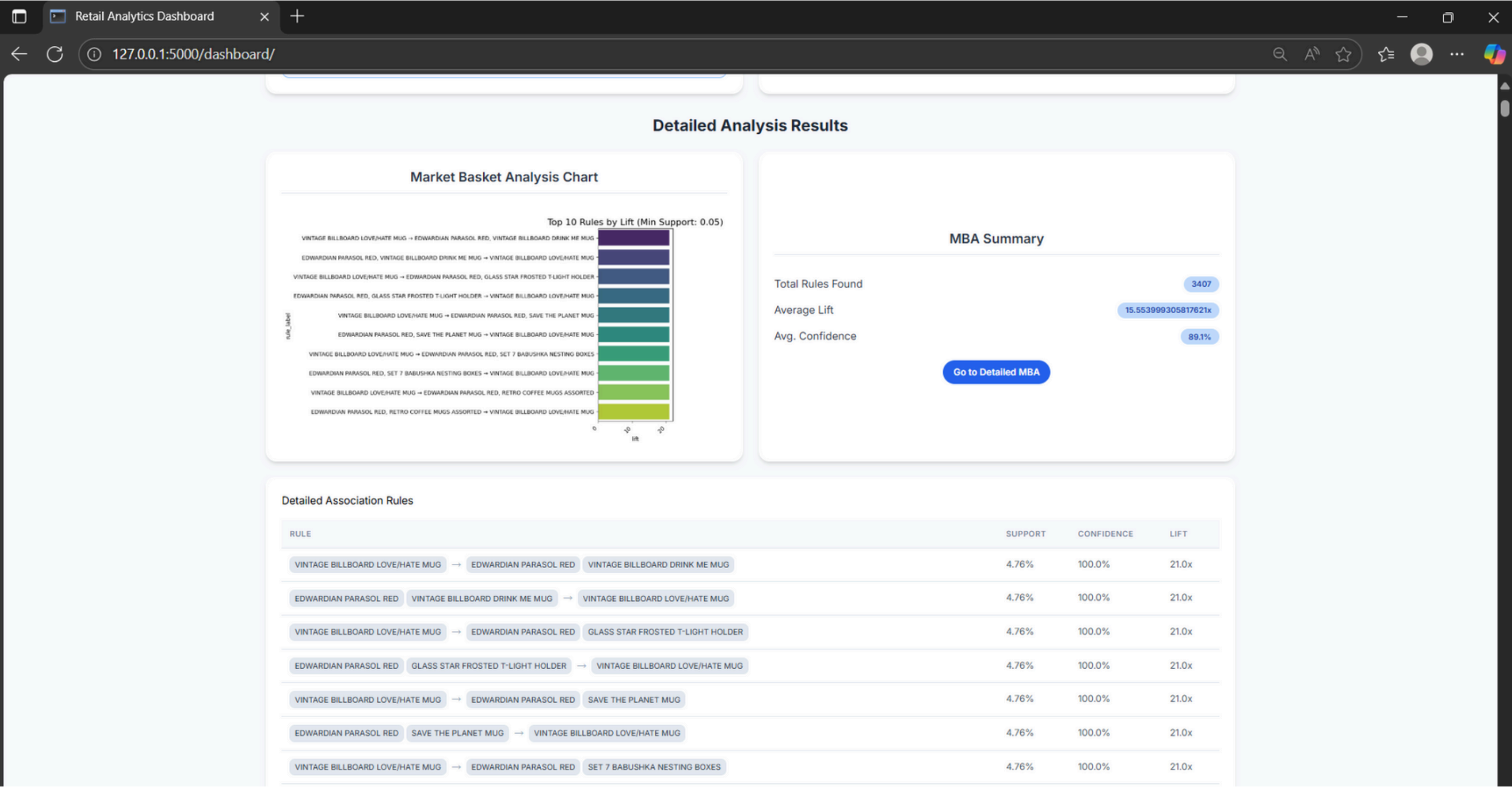
Next, Buy: RETROSPOT HEART HOT WATER BOTTLE

Popular sequential purchase recommendation based on customer patterns.

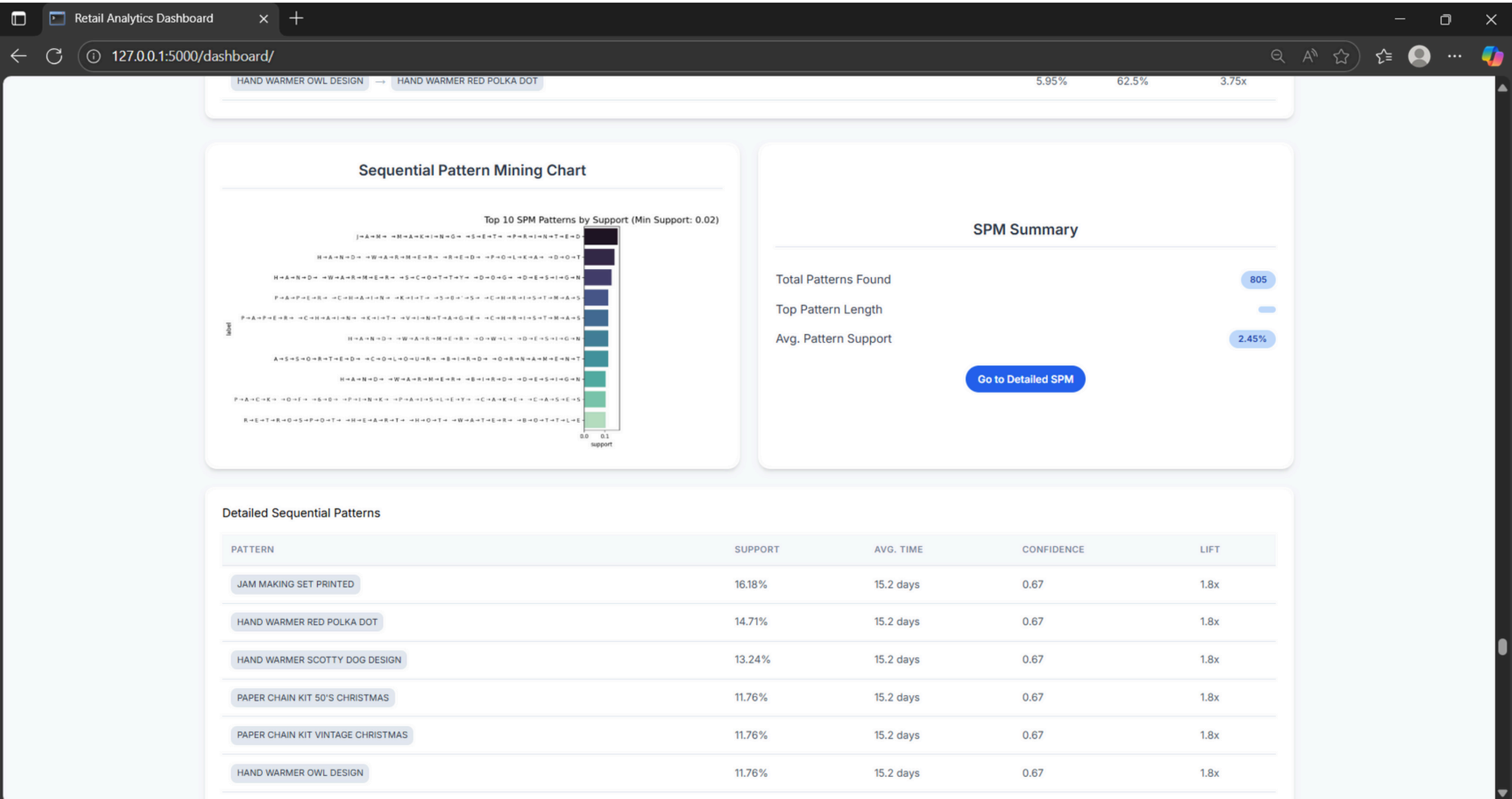
CHRIST (Deemed to be University) *Excellence & Service*

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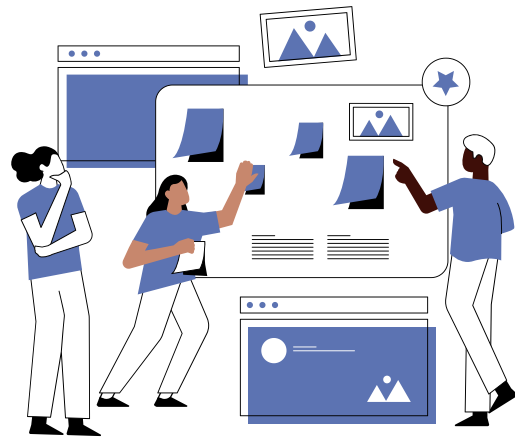
Demo: Bundle Recommendations Tab (Market Basket Analysis (MBA) Insights)



Demo: Bundle Recommendations Tab (Sequential Pattern Mining (SPM) Insights)



Value Delivered So Far



- **Actionable Intelligence:** Retailers can now understand what products are truly associated and how customers navigate their purchases.
- **Empowered Decision-Making:** Directly informs cross-selling campaigns, product placement, and customer journey optimization.
- **Simplicity:** Advanced analytics made accessible to non-technical users.
- **Robust Foundation:** A scalable platform for further analytical expansion.

Future Steps: Expanding the Dashboard (Phase 2)

- **Goal:** Integrate broader business insights into the central dashboard for a truly comprehensive view.



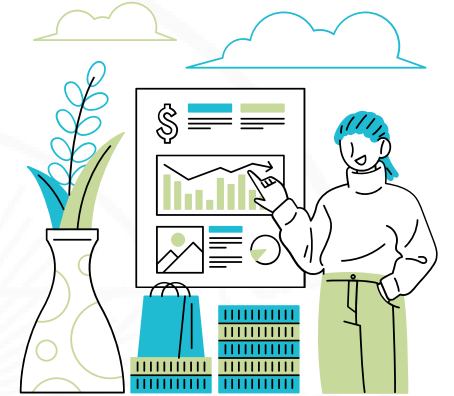
- **Sales Trends & Historical Analysis:**

- Visualize historical sales performance (revenue, quantity over time).
- Identify seasonal patterns, growth trajectories, and anomalies.
- Allow filtering by product, category, or customer segment.

Future Steps: Expanding the Dashboard (Phase 2)

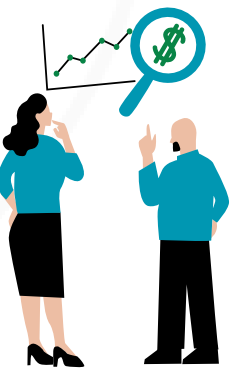
- **Future Sales Forecasting:**

- Develop and integrate predictive models (e.g., ARIMA, Prophet) to forecast upcoming sales volumes for overall business or specific products.
- Assist with inventory management, staffing, and purchasing decisions.



- **Pricing Optimization Models:**

- Analyze demand elasticity (how sensitive sales are to price changes).
- Implement optimization algorithms to suggest optimal pricing for individual products and bundles to maximize revenue or profit.



Future Steps: Expanding the Dashboard (Phase 2)



- **Integrated Recommendation Engine:**
 - Combine insights from MBA, SPM, Sales Trends, and Pricing Optimization to formulate dynamic, personalized product and bundle recommendations that include suggested pricing.
 - Move beyond just identifying patterns to actively suggesting "Buy this bundle at this price for maximum impact."





Summary & Conclusion



- We've built a strong foundation for a user-friendly, powerful Retail Sales Analytics Platform.
- Our vision is to empower small and mid-size retailers with sophisticated analytics that are simple to use and directly drive business growth.
- We are poised for the next phase of integrating predictive and prescriptive insights.



THANK YOU!

